

# 1.1.4 – 1.1.6 Boolean Algebra, Operations & Truth Tables

بولین الجبرا جارج بول (George Boole) نے بنایا۔

CS XI

Chapter 1: Computer Systems

Lecture 03 of 12

## 1.1.4 – 1.1.6 Boolean Algebra, Operations & Truth Tables

### LEARN IT

**Boolean algebra** was developed by the English mathematician **George Boole**. It is a kind of algebra that deals with only two values: **TRUE (1)** and **FALSE (0)**. It is the foundation of digital electronics and computer programming — every `if` condition in a program is a small piece of Boolean algebra.

| Concept     | Description                               | Example              |
|-------------|-------------------------------------------|----------------------|
| Variables   | Letters that hold the value 0 or 1.       | A, B, C, X, Y        |
| Operations  | Actions performed on variables.           | AND, OR, NOT         |
| Expressions | Combinations of variables and operations. | A AND B, $Y = A + B$ |

Some simple rules (laws) follow directly from the meaning of AND, OR and NOT. They are useful when you check or simplify answers:

| OR rules    | AND rules       | NOT rules        |
|-------------|-----------------|------------------|
| $A + 0 = A$ | $A \cdot 1 = A$ | $(A')' = A$      |
| $A + 1 = 1$ | $A \cdot 0 = 0$ | $A + A' = 1$     |
| $A + A = A$ | $A \cdot A = A$ | $A \cdot A' = 0$ |

### WORKED EXAMPLE

**Using a law:** Simplify  $Y = A'B + AB$ .

Step 1: Both terms contain B, so take B out:  $Y = B(A' + A)$ .

Step 2:  $A' + A = 1$  (a variable OR its opposite is always 1).

Step 3:  $Y = B \cdot 1 = B$ .

### COMMON MISTAKE

In Boolean algebra "+" means OR, not addition. So  $1 + 1 = 1$  (TRUE OR TRUE is TRUE), not 2. There is no value bigger than 1.

### اردو میں سمجھیں

بولین الجبرا جارج بول (George Boole) نے بنایا۔ اس میں صرف دو قیمتیں ہوتی ہیں: درست (1) اور غلط (0)۔ متغیرات (Variables) جیسے A اور B صرف 0 یا 1 رکھتے ہیں، آپریشنز AND، OR اور NOT ہیں، اور ان کے ملاپ سے ایکسپریشن (Expression) بنتا ہے۔

### CHECK YOURSELF

1. Who developed Boolean algebra?
2. In Boolean algebra, what is  $1 + 1$ ?

Answers: 1. George Boole. 2. 1 (because + means OR).

### LEARN IT

There are three basic Boolean operations. Every other operation can be made from them.

1. **AND operation (  $\cdot$  )** — gives output 1 **only when ALL inputs are 1**. Expression:  $Y = A \cdot B$  (often written just AB).
2. **OR operation ( + )** — gives output 1 **when ANY input is 1**. Expression:  $Y = A + B$ .
3. **NOT operation ( ' or  $\bar{\phantom{x}}$  )** — **inverts** (flips) the input. If the input is 1 the output is 0, and if the input is 0 the output is 1. It is a **unary** operation, meaning it works on only one operand. Expression:  $Y = A'$  (also written as A with a bar on top,  $\bar{A}$ ).

| Operation | Real-life analogy                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------------------------|
| AND       | A door with two locks. The door opens only when <b>both</b> locks are unlocked.                                                    |
| OR        | A room with two doors. You can enter if <b>either</b> door is open.                                                                |
| NOT       | A room light switch. When the switch is ON the room is lit; when it is OFF the room is dark — the state is the opposite of "dark". |

### WORKED EXAMPLE

Evaluate each expression for  $A = 1, B = 0$ .

- (a)  $A \cdot B = 1 \cdot 0 = 0$  (not all inputs are 1).
- (b)  $A + B = 1 + 0 = 1$  (at least one input is 1).
- (c)  $B' = 0' = 1$  (NOT flips 0 to 1).
- (d)  $A \cdot B' = 1 \cdot 1 = 1$ .

### REMEMBER IT

AND is "strict" — everyone must agree (all 1s). OR is "easy-going" — one 1 is enough.  
NOT is "the rebel" — it always says the opposite.

### اردو میں سمجھیں

AND آپریشن تب ہی 1 دیتا ہے جب تمام ان پٹ 1 ہوں (دو تالوں والا دروازہ). OR آپریشن تب 1 دیتا ہے جب کوئی ایک ان پٹ بھی 1 ہو (دو دروازوں والا کمرہ). NOT آپریشن ان پٹ کو الٹ دیتا ہے اور صرف ایک ان پٹ پر کام کرتا ہے، اس لیے اسے یونری (Unary) کہتے ہیں۔

### CHECK YOURSELF

1. What symbol is used for the OR operation?
2. Why is NOT called a unary operation?
3. Find  $A + B'$  when  $A = 0, B = 1$ .

Answers: 1. "+". 2. Because it works on only one operand (one input). 3.  $B' = 0$ , so  $0 + 0 = 0$ .

### LEARN IT

A **truth table** shows **all possible input combinations** and the output for each one, for a logic gate or a circuit. It is the "complete record" of how the circuit behaves.

#### How to create a truth table:

1. Count the number of inputs ( $n$ ).
2. Calculate total rows =  $2^n$ .
3. List all input combinations in binary counting order (0, 1, 2, 3 ... in binary).
4. Calculate the output for each combination.

#### 2-input example ( $n = 2$ , rows = 4)

| A | B | Combination |
|---|---|-------------|
|---|---|-------------|

#### 3-input example ( $n = 3$ , rows = 8)

| A | B | C | Decimal |
|---|---|---|---------|
|---|---|---|---------|

|   |   |       |
|---|---|-------|
| 0 | 0 | Row 1 |
| 0 | 1 | Row 2 |
| 1 | 0 | Row 3 |
| 1 | 1 | Row 4 |

|   |   |   |   |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 2 |
| 0 | 1 | 1 | 3 |
| 1 | 0 | 0 | 4 |
| 1 | 0 | 1 | 5 |
| 1 | 1 | 0 | 6 |
| 1 | 1 | 1 | 7 |

#### REMEMBER IT

**Quick fill trick for 3 inputs:** In column C write 0,1,0,1... (change every row). In column B write 0,0,1,1... (change every 2 rows). In column A write 0,0,0,0,1,1,1,1 (change every 4 rows). For n inputs the leftmost column changes every  $2^{n-1}$  rows.

#### WORKED EXAMPLE

**Make the truth table of a 3-input AND:  $Y = A \cdot B \cdot C$ .**

Step 1:  $n = 3$ , so rows =  $2^3 = 8$ .

Step 2: List the 8 combinations in binary order.

Step 3: AND gives 1 only when all inputs are 1, so only row 111 gives 1.

| A | B | C | $Y = A \cdot B \cdot C$ |
|---|---|---|-------------------------|
| 0 | 0 | 0 | 0                       |
| 0 | 0 | 1 | 0                       |
| 0 | 1 | 0 | 0                       |
| 0 | 1 | 1 | 0                       |
| 1 | 0 | 0 | 0                       |
| 1 | 0 | 1 | 0                       |
| 1 | 1 | 0 | 0                       |
| 1 | 1 | 1 | 1                       |

### COMMON MISTAKE

Do not list rows in a random order — you may miss or repeat a combination. Always count in binary from 000 to 111. Also remember rows =  $2^n$ , not  $2 \times n$ : 3 inputs give 8 rows, not 6.

### اردو میں سمجھیں

ٹرو تھ ٹیبل (Truth Table) کسی گیٹ یا سرکٹ کے تمام ممکنہ ان پٹ اور ان کے آؤٹ پٹ دکھاتا ہے۔ اگر ان پٹ  $n$  ہوں تو قطاریں (Rows) 2 کی طاقت  $n$  ہوں گی؛ دو ان پٹ کے لیے 4 اور تین ان پٹ کے لیے 8 قطاریں۔ ان پٹ ہمیشہ بائری ترتیب میں لکھیں۔

### CHECK YOURSELF

1. How many rows does a truth table with 4 inputs have?
2. In a 3-input table, what is the 6th row (counting from row 1 = 000)?

Answers: 1.  $2^4 = 16$ . 2. 101.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** Boolean algebra uses:

- a. TRUE(1) and FALSE(0) only
- b. All real numbers
- c. ASCII letters
- d. IPv4

**MCQ 2.** AND is 1 only when:

- a. Every input is 1
- b. Any input is 1
- c. Inputs differ
- d. There is no input

**MCQ 3.** OR is 1 when:

- a. Any input is 1
- b. All inputs are 0
- c. Inputs are equal
- d. NOT is applied twice

**MCQ 4.** NOT of 1 is:

- a. 0
- b. 1
- c. 2
- d. A

**MCQ 5.** Rows in an n-input truth table:

- a.  $2^n$
- b. n
- c.  $n^2$
- d.  $n+1$

**MCQ 6.** Two-input truth table has:

- a. 4 rows
- b. 2 rows
- c. 8 rows
- d. 16 rows

## Extra short questions

1. Who developed Boolean algebra and what two values does it use?
2. Write  $Y = A \cdot B$  and  $Y = A + B$  in words.
3. Why is NOT called unary?
4. List the four steps to build a truth table.

## Extra long question

1. Write complete truth tables for 2-input AND, OR and 1-input NOT. Add one real-life lock/door/switch example each.

## Extra answer key

**MCQs:** 1. A 2. A 3. A 4. A 5. A 6. A

### Short

1. George Boole. TRUE(1) and FALSE(0).

2. AND: output 1 only if all inputs are 1. OR: output 1 if any input is 1.
3. It has only one operand: it inverts a single input.
4. Count inputs  $n$ ; rows =  $2^n$ ; list combinations in binary order; compute each output.

**Long**

1. AND 0001 on AB=11 (two locks). OR 0111 (two doors). NOT 10 / 01 (light switch). Show all four/two rows.

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Next lecture: 1.1.7 Basic Logic Gates (AND, OR, NOT)